Automated Hybrid Stair Climber for Physically Challenged

1st Preeta Chatterjee
Dept. of ECE, IEM
Institute of Engineering and Management
Kolkata, India

2nd Neelakshi Lahiri
Dept. of ECE, IEM
Institute of Engineering and Management
Kolkata, India

3rd Anirban Bhattacharjee
Dept. of ECE, IEM
Institute of Engineering and Management
Kolkata, India

4th Arindam Chakraborty
Dept. of ECE, IEM
Institute of Engineering and Management
Kolkata, India

***Abstract*—Ascending stairs has always been a trouble for the disabled, especially for people with neurological disorders. Wheelchairs have been developed over time and now some even have the ability to climb stairs. In this paper, several such devices like SCEWO and the novel stair climbing wheelchair have been surveyed. However, many come with the disadvantage of being either bulky or extremely costly. Since people in India come from different financial backgrounds, there should be a device which helps the ones in need without thrusting even more problems in the form of extra costs on them. Yet, the device should be stable enough for them to use safely, even at a lower cost. So, by combining the different functionalities of the current devices such as wheel clusters, tracks, and rotating legs, and adding some more of our own, our aim is to design a cost-effective, stable, hybrid stair climbing device for neurological patients.**

***Index Terms*—ascending stairs, neurological disorders, cost-effective, hybrid stair climbing device**

I. Introduction

The wheelchair was first invented in the 17th Century, by a German boy to get over his disability, but stairs were invented even before that. Thus a new problem arose that the wheelchair could not be used on stairs. With the advancement of technology, inventions have been made to solve that problem too. The first wheelchair with the ability to climb stairs was invented by Ernesto Blanco in 1962.

As seen in figure 1, the wheelchair is space consuming. It is foldable, but would still have a width of 7 inches and would weigh 80 pounds [1]. It has a limitation in the usage in the staircase, that is, it can only be used in staircases wider than 32 inches. Being the first of its kind, this wheelchair came with many drawbacks. Many more models of stair climbing wheelchairs were developed after that, but they always came with some disadvantage or the other.

Another wheelchair employing an idea similar to the one previously discussed was invented in 2013. This wheelchair uses a Penta wheel and is more economical and was made with the motive to be easily accessible to middle-class people [2].

This device also came with a disadvantage. To quote the inventors, "while coming down from the steps, we found some vibration problems". Apart from that, this device is also space-consuming and must be manually operated thereby limiting its usage.

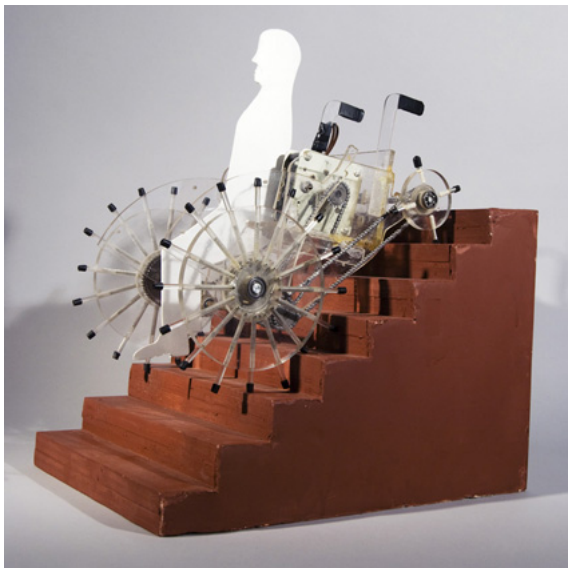

Fig. 1. Staircase Climbing device developed by Ernesto Blanco [3]

Another idea to make wheelchairs functional on stairs has been employed. This wheelchair uses a belt system as in army tanks and heavy machines to run over rough terrains.

Here (Fig 2), the use of the belt limits the usage of the device. Also, due to its size, it cannot be used on all stairs and is not very convenient to carry.

Wheelchairs in themselves are bulky - the lightest automatic wheelchair weighs approximately 15 kgs. This affects their portability. On adding the stair climbing facility, both their dimensions and weight increase. To reduce those factors, the idea of a device, which isn't usually used to carry a human being, can be employed and modified to suit the needs of humans with disabilities, especially with neurological disorders to climb stairs. This will reduce the size, weight and limitations due to staircase widths.

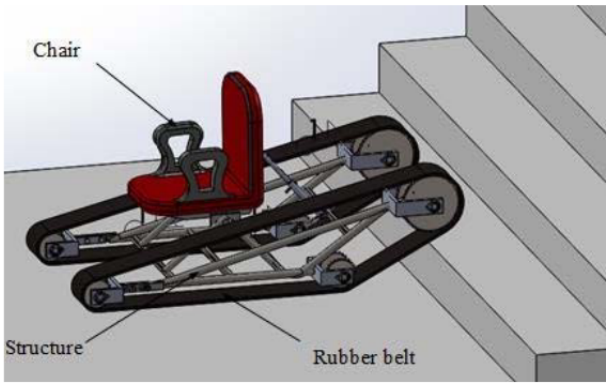

Fig. 2. Rubber Belt Wheelchair [4]

In this paper, some existing devices from which ideas can be gathered to make a functional stair climbing device with the ability to carry humans have been surveyed.

II. SURVEY ON PRESENT STAIR CLIMBING DEVICES

Through the course of developing a device, the stair climbing mechanism has to be designed suitably. The design itself would directly affect the weight, cost, energy consumption, portability, and ease of use. The basic mechanisms can be classified into four main categories [5] :

A. Track-based stair climbing system

Tracks generally have ridges that press on the edges of the stairs and the motor-driven mechanism helps it rotate along the inner surface of the track [6]. This makes the device move up and down the stairs.

B. Wheel cluster-based stair climbing system

A component having multiple wheels that are uniformly distributed around a common centre is called a wheel cluster. The number of wheels might range from two to four, but can be higher too [7]. For stair climbing, the wheels rotate around the central axis and that force propels the device up and down the stairs.

C. Leg-based stair-climbing mechanism

These devices are created from the idea of imitating human and animal leg movements, and stair climbing techniques. These devices use leg-like contraptions to aid stair climbing and make it one of the more efficient and stable methods [9].

D. Hybrid stair-climbing mechanism

Such devices usually use a combination of the different mechanisms to figure out a more economical solution. They are more innovative and have a more complex overall structure [8].

Below we have surveyed some machines which have been built with different aims but the same goal - to aid people to climb stairs.

A. SCEWO

SCEWO is a balanced wheelchair that can move up and down stairs independently and smoothly. This innovation gives its users a powerful experience of walking over obstacles, with many rubber tracks providing smooth and safe staircases.

Fig. 3. SCEWO is a balanced wheelchair [10]

This device provides different modes for different functionalities. The safe mode is used for entry and exit, table access, or standing. The high mode enables eye-to-eye contact and access to higher objects. The track mode overcomes slippery surfaces such as snow or gravel, tricky obstacles, or steeper slopes. To help the user overcome stairs independently without an external power source, a manual mode is added too.

B. Novel stair-climbing wheelchair

The main aim of building this wheelchair was to make use of the power of the now-defunct body, by using rotating legs driven by a rotating position. This device doesn't use expensive heavy metals or electric engines, thereby making it economical. An upgraded version consisting of hand-wheels provides power equal to a standard wheelchair, with the added benefit of being compact and lightweight. The wheelchair has a rotating leg machine based on lever propulsion control for climbing stairs with a height of 160 mm [11].

Fig. 4. Novel stair-climbing wheelchair operated manually [11]

III. PROPOSAL FOR A HYBRID STAIR CLIMBING DEVICE

Physical disability is a problem people end up facing if they live long, as bone strength and muscles deteriorate. Also, unfortunately, many people have to face it due to accidents,

birth defects, or illnesses such as neurological problems. People with such disabilities usually choose a wheelchair for mobility inside and outside of the home. Wheelchairs have been evolving for a long time. However, they face a lot of difficulties trying to climb stairs, and many a-times, it becomes impossible for them. As an alternative, people usually carry the person in a wheelchair as in the figure, with one person on each side, or carry the elderly or disabled on their backs [12]. This method may be the easiest option available but comes with a horde of risks with the main one being a risk of injury for both the carrier and the person being carried.

Fig. 5. Common technique for climbing stairs [13]

Here in India, there are many people who live in their ancestral homes who come from different financial backgrounds. Several more people live in multi-storeyed buildings which do not have the provision for constructing an elevator, leaving the stairs to be the sole passage for traversing to and from their house. Among these people, the ones who have disabilities of any sort face lots of problems in climbing up and downstairs.

Pre-existing devices come with many disadvantages. They are often bulky and heavy thereby reducing their portability. Some like the track based devices may cause damage to the stairs as the tracks push against the stairs heavily. Apart from these, the devices may also be breakable, difficult to operate and yet be expensive.

To provide people with stair climbing devices to aid them to traverse, which is as economic for them as it is useful, seems to be the best way to tackle this problem.

IV. HYPOTHETICAL IMPLEMENTATION FOR OUR PROPOSED IDEA

Our objective is to put together a hybrid stair climbing machine, by modifying prevailing devices. Currently available devices either use a wheel cluster to ascend up the stairs which is operated manually, a track-based device that applies pressure on the stairs to climb up, or a rotating leg operated by motors. So, in our device, we want to use both a rotating leg and a wheel cluster to automate the device as well as maintain stability.

Fig.6 shows the rough working of the model we are hoping to develop. It will have a Primary DC Source which will power 2 DC motor units. The Main DC motor unit will power the main axle which will rotate both the climbing rotor blades. The Auxiliary DC motor unit will power a pair of triangular tri-wheel clusters connected to a supporting axle to support the function of Main Axle.

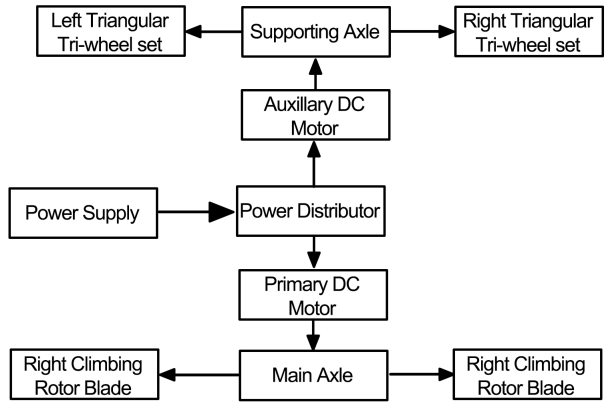

Fig. 6. A schematic diagram of our proposal

V. CONCLUSION AND FUTURE SCOPE

Starting with the wheelchair, many devices have been made for aiding people to move. A stair-climbing device, whether in the form of a wheelchair or a different device altogether, provides convenient and effective assistance to people.

We have proposed a hybrid stair-climbing device in our article. They are relatively new and may have limited reviews, but if formulated correctly, they have the ability to be the very thing needed by the people. This device should be portable, less bulky, cost-effective, automatic and simple to use. All these disadvantages in the available devices provide a wide avenue for future improvisations.

With the advance of technology, the aim of engineers all over the world is to enhance prevalent devices. With a similar aim, this survey has been put together for the betterment of mankind.

REFERENCES

[1] E. E. Blanco, Stair Climbing Wheelchair - Case Study in Creative Design
[2] R Rajasekar, K P Pranavkarthik, R Prashanth, S Senthil Kumar and A Sivakuma, April 2013. Design and Fabrication of Staircase Climbing Wheelchair. International Journal of Mechanical Engineering and Robotic Research, Vol. 2, No. 2.
[3] http://museum.mit.edu/150/17
[4] Maruti Khot, Anurag Gumaste, Varsha Jagdale, 2016. Design And Development Of Step Climbing Wheelchair. International Journal Of Advances In Production And Mechanical Engineering, Vol. 2, Issue 5.
[5] Weijun Tao, Junyi Xu and Tao Liu. Electric-powered wheelchair with stair-climbing ability, International Journal of Advanced Robotic Systems, July-August 2017: 1–13
[6] Weijun Tao, Yi Ou and Hutian Feng - Research on Dynamics and Stability in the Stairs-climbing of a Tracked Mobile Robot, International Journal of Advanced Robotic Systems, Vol. 9, 2012
[7] Tumula Mani Kota Rajasekhar, M.Sugadev - Arduino Controlled Special Stair Climbing Wheelchair Bot, International Journal of Pure and Applied Mathematics, Vol. 118, No. 24, 2018
[8] Nayan Jyoti Baishya and Harutoshi Ogai - Wa-Chair: A concept for development of economical stair-climbing wheelchairs, IOP Conf. Series: Materials Science and Engineering, 307, 2018
[9] Rupesh Gupta, Chitkara University, International Journal of Scientific Research, Vol. 4, Issue 10.

[10] SCEWO, URL: http://scalevo.ch/
[11] Murray J. Lawn, Takakazu Ishimatsu - Modeling of a Stair-Climbing Wheelchair Mechanism With High Single-Step Capability, IEEE Transactions on Neural Systems and Rehabilitation Engineering, Vol. 11, No. 3, September 2003
[12] Kai Sasaki, Yosuke Eguchi, Kenji Suzuki - Stair-climbing wheelchair with lever propulsion control of rotary legs
[13] https://www.miusa.org/